

Project Notes and Documentation

Enterprise AI Tutor is a modular AI-powered learning platform built with Python, Streamlit, MySQL, FAISS, and multiple AI providers.

Objectives: Build a scalable AI learning platform with Enterprise RAG, Voice AI, modular architecture, and comprehensive documentation.

Architecture: Streamlit UI → Service Layer → Provider Layer → AI Models → MySQL & FAISS.

Technology Stack: Python, Streamlit, MySQL, FAISS, Ollama, OpenAI, Anthropic, Faster Whisper, gTTS.

Major Features: AI Tutor, Enterprise RAG, Voice AI, PDF Tutor, Coding Tutor, AI Mentor, Study Planner, Quiz Generator, Flashcards, Notes Generator, Analytics.

Engineering Practices: OOP, Factory Pattern, Strategy Pattern, Layered Architecture, Separation of Concerns, High Cohesion, Loose Coupling, Logging, Exception Handling.

Documentation: README, Architecture, System Design, Database, RAG, Voice AI, Provider Guide, Installation, User Guide, API Docs, Security, FAQ, Troubleshooting, Interview Guide, Resume Bullets.

Outcome: Demonstrates end-to-end AI engineering, software architecture, enterprise RAG, Voice AI, and technical documentation.